

Color



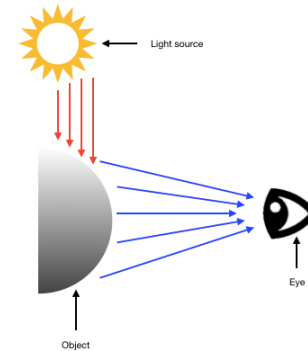
Basic color theory

Color perception depends on:

- Nature of light
- Interaction between light and matter
- Physiology of human vision

Essential elements to deal with color:

- Light
- Object
- Observer



Some physics

Color physics...

Light is an electromagnetic wave

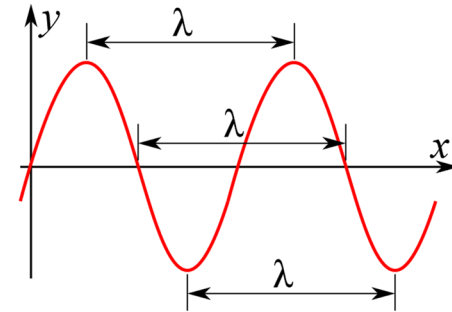
That is, a kind of energy: any color has a corresponding *wavelength*

λ = *wavelength*

T = *period* (time between two consecutive transits of a “crest” through a specific point)

ν = *frequency* = $1/T$ (how many times, in a second, a crest passes through a specific point)

v = *speed* = $\lambda/T = \lambda\nu$ = distance traveled by the wave in a second ($\sim 3 \cdot 10^8$ m/s)



Since the speed is constant, λ and ν are in inverse proportion

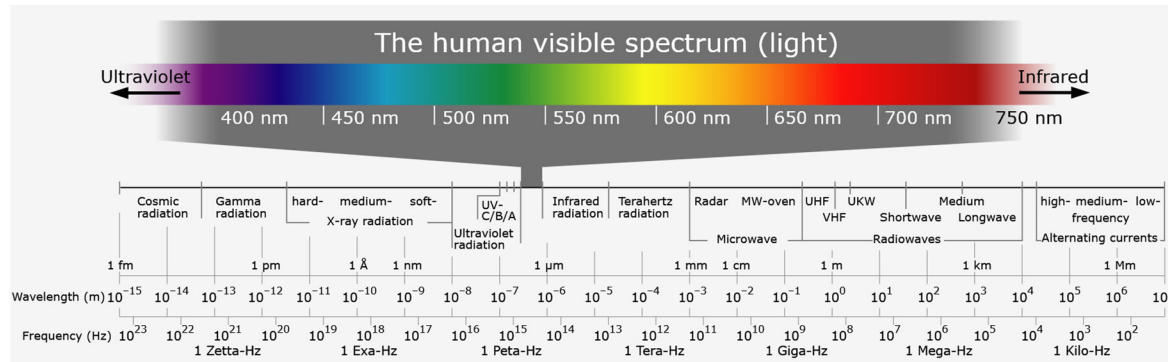
Some physics

... color physics...

All possible wavelengths form the *electromagnetic spectrum*

The human eye can perceive only some wavelengths

Those in the interval 380-780 nm, which form the *visible spectrum*



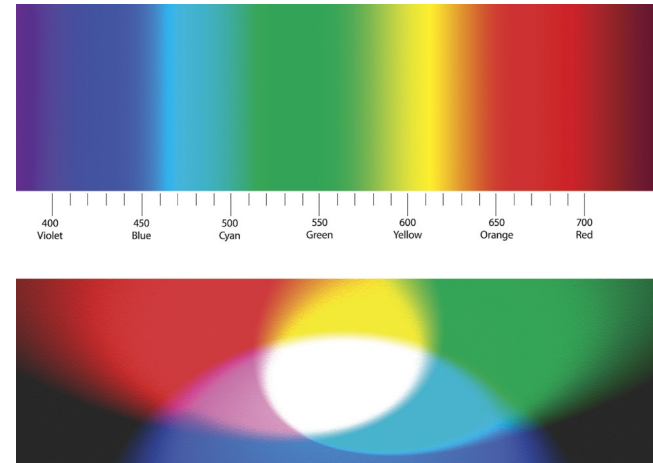
Some physics

... color physics...

All wavelengths of the visible spectrum, together, generate *white light*

The color of an object depends on the kind of light under which it is observed. E.g.:

- Incandescent lamp
- Neon
- Gas discharge
- Led
- ...



Some physics

... color physics...

Color temperature

- Heat from a light source (measured in Kelvin degrees)
- Under a “warm” light, colors look more vivid than under a “cool” light
E.g.: under a cool light, blues are darker, greens tend to yellow, and pinks tend to red

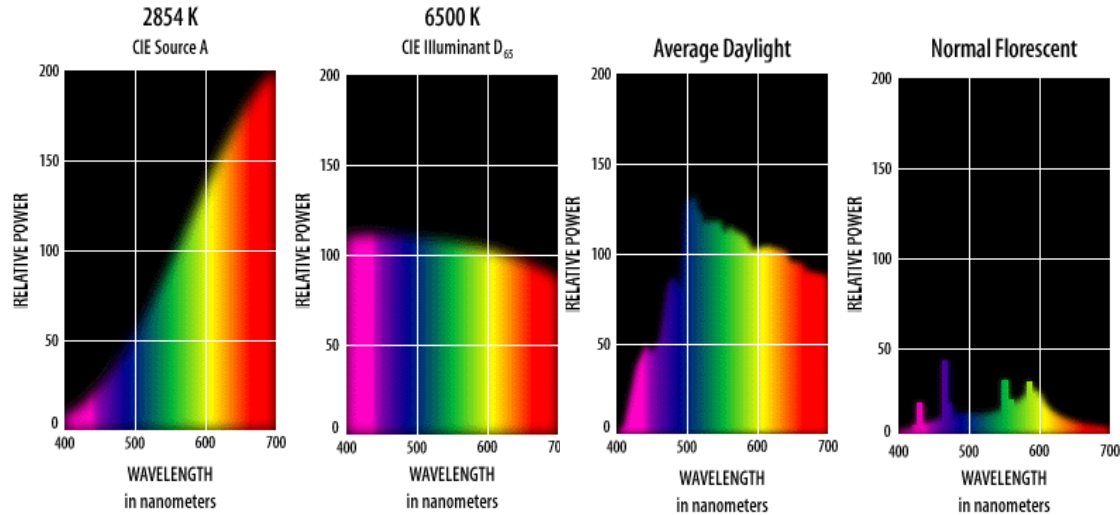
Spectral Power Distribution

Refers to wavelengths corresponding to the light emitted by a source with a specific color temperature

Some physics

... color physics...

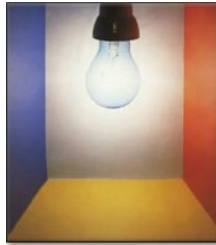
Examples of spectral power distributions



Some physics

... color physics

Examples of different lighting effects with different sources



Virtual lamp
(natural light)



Mercury lamp



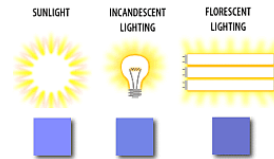
High
pressure
sodium lamp



Incandescence
lamp



Low pressure
sodium lamp



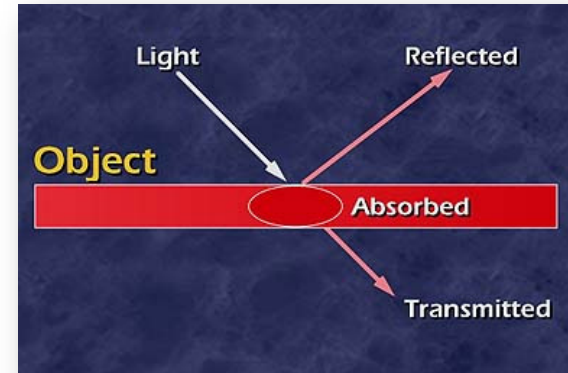
Light and its “behavior”

Interaction between light and matter...

When light hits an object, its wavelengths can be *reflected*, *absorbed*, or *transmitted* (or a combination of these)

It all depends on the object's surface

Transparent, opaque, shiny, ...

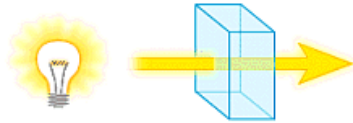


... interaction between light and matter...

Transmission

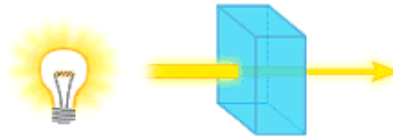
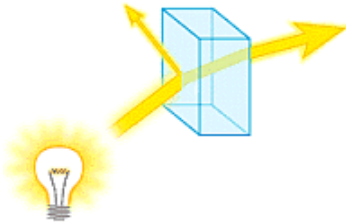
- Total

Transparent object and perpendicular incidence



- Partial

Transparent object and non-perpendicular incidence or translucent object

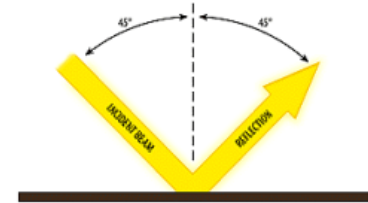


Light and its “behavior”

... interaction between light and matter...

Reflection

Opaque object with shiny surface



Diffusion

Opaque object with coarse surface



Reflection/diffusion

Opaque object neither shiny nor coarse
(most common case)



... interaction between light and matter...

Absorption

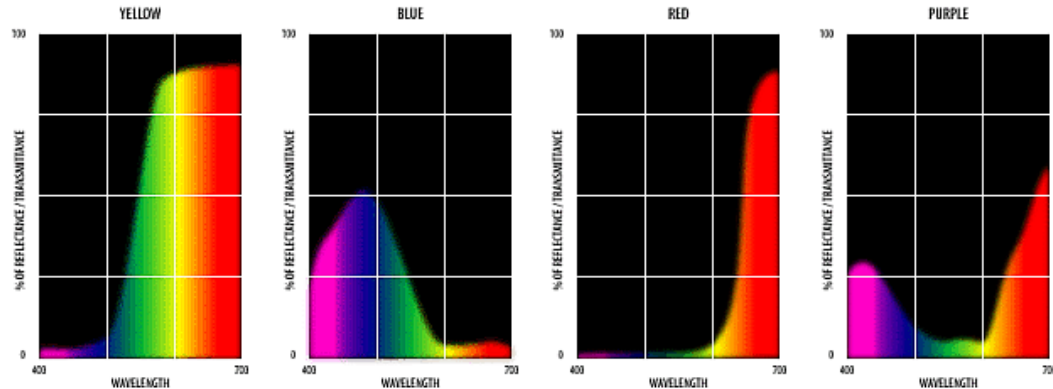
- Depends on the pigmentation of the object (natural or artificial color)
- Only some wavelengths are absorbed, while the others are reflected
The perceived color is given by the wavelengths that are not absorbed



... interaction between light and matter

Spectral Reflectance/Transmittance curve

For each wavelength, it represents the light reflected by an opaque colored object (compared to pure reflection) or transmitted through a colored transparent object (compared to a completely transparent material)



Physiology of human vision...

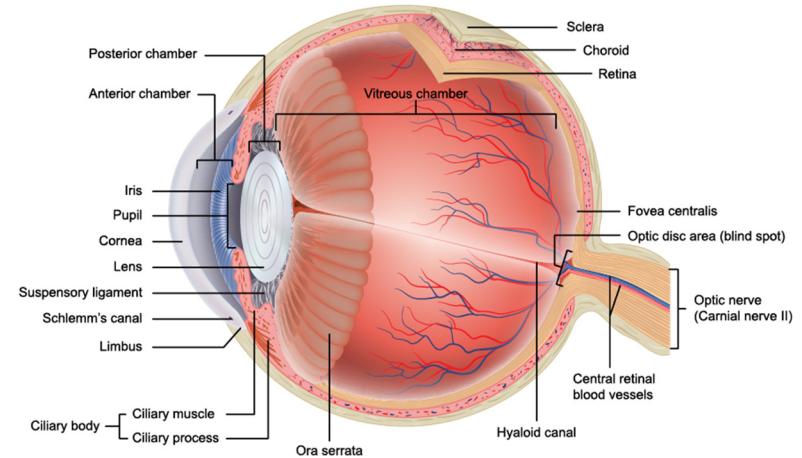
Rays of light enter the *cornea* and, through the *pupil*, are focused on the *lens*

The size of the pupil is modified by the *iris*, depending on the amount of light

From the lens, through the *vitreous humor*, light is transmitted to the *retina*

Pictures are focused on the retina upside down

Perceived by *photoreceptors* of the retina, light is delivered to the brain by means of the *optic nerve*



There are two kinds of photoreceptors: *cones* and *rods*

... physiology of human vision...

Cones and rods are especially concentrated in the *fovea*

The fovea is the part of the retina more sensitive to light and colors

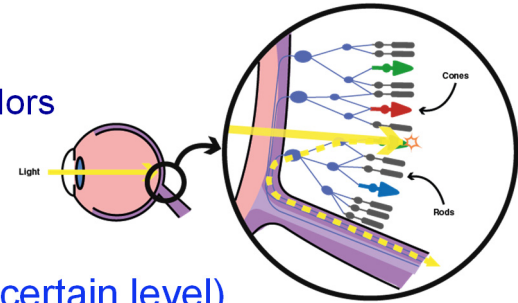
Rods are sensitive to light but not to colors

Cones are sensitive to colors but insensitive to light (under a certain level)

Some cones are sensitive to red wavelengths (ρ cones), others to green wavelengths (γ cones), and others to blue wavelengths (β cones)

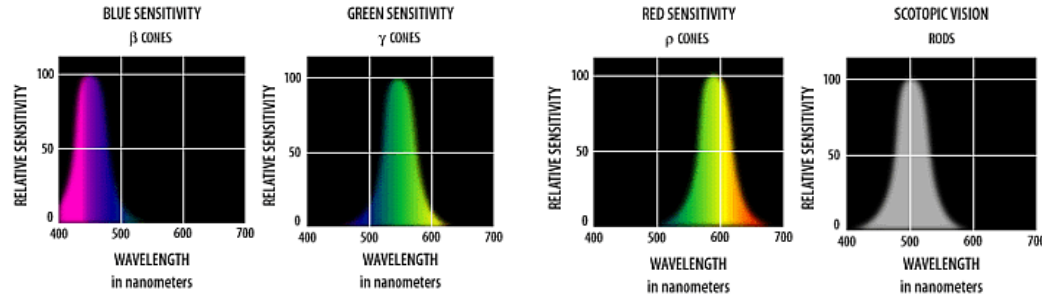
Spectral sensitivity

Sensitivity level of cones and rods with different wavelengths



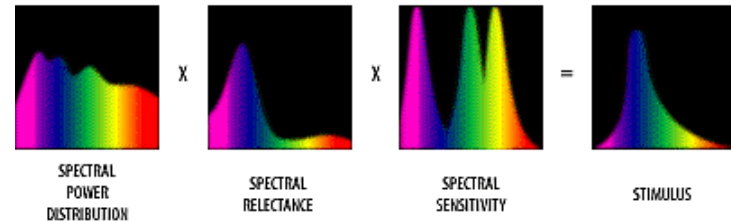
... physiology of human vision

Spectral sensitivity



Color stimulus delivered to the brain

Depends on: (a) illumination, (b) object pigmentation, and (c) eye sensitivity



Subjective aspects of color perception

Different people may receive different stimuli from colors

Color is often a “personal experience”...

“Warm” and “cool” colors

Due to our mental associations

Yellow, orange, red → sun, fire, ...

Green, blue, purple → plants, sky, sea, ...

Warm colors are considered “centripetal”, while
cool colors are considered “centrifugal”

The observer perceives them as “closer” or “farther”



Color models

Used to classify and characterize colors, based on their features, digitally

Main models:

- *RGB* (additive)
- *CMY(K)* (subtractive)
- *HSB/HSV*
- *CIE (CIE XYZ, CIE LUV, CIE L^*a^*b)*

Photo editing software tools (e.g., Adobe Photoshop or GIMP) can manage different color models

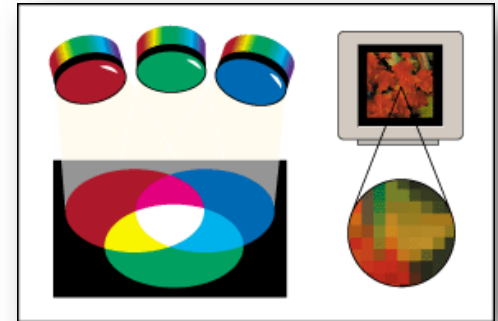
RGB model...

RGB = Red Green Blue

- Used in screens, video projectors, ambient lighting, scanners
- A large part of the visible spectrum can be obtained by mixing different amounts of the three basic colors: *red*, *green*, and *blue* (values from 0 to 255 or from 0% to 100%)

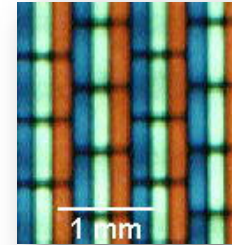
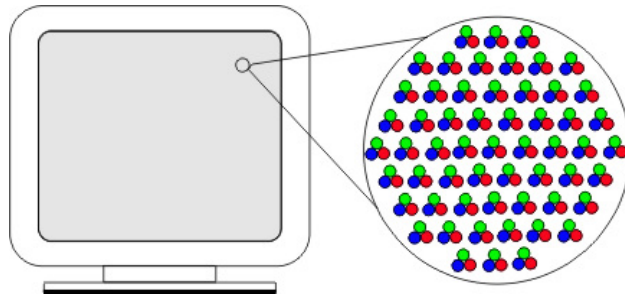
Additive model

- Mixing the three colors at their maximum intensities, we obtain white (all the light is reflected)
- The combinations of pairs of colors produce cyan, magenta, and yellow



Color coding

... RGB model



CMY(K) model...

CMY(K) = Cyan Magenta Yellow (*black*K)

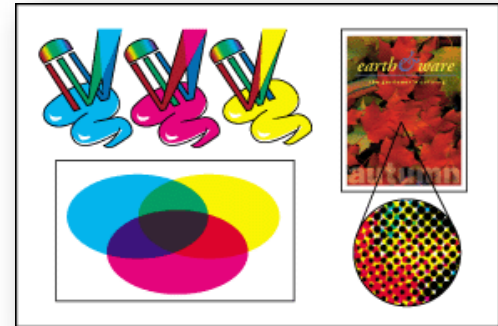
Used in printing, based on the ability of inks to absorb light on paper (when white light hits translucent inks, a part of the spectrum is absorbed and a part is reflected)

Subtractive model

Values from 0% to 100%; white is obtained with cyan, magenta, and yellow at level 0

But *black* is always necessary...

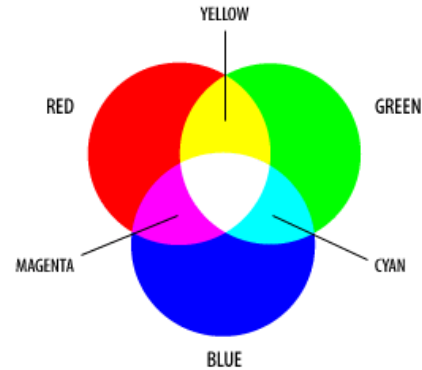
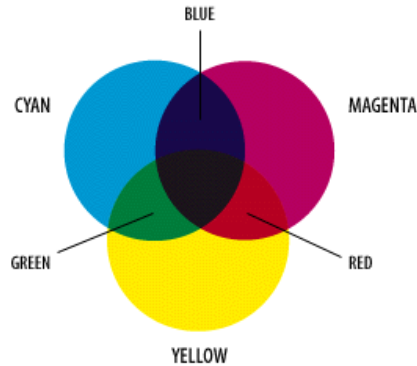
The CMY and RGB models are *complementary*



Color coding

... CMY(K) model...

CMY-RGB complementarity

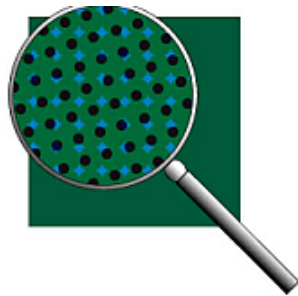
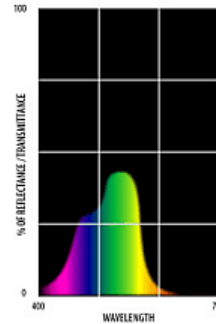
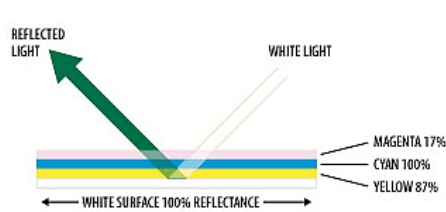


Color coding

... CMY(K) model

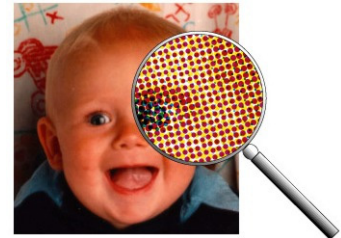
Printing process

Overlapped layers of cyan, magenta, and yellow inks



Halftoning gives the illusion of "continuity"

Black is always necessary



RGB and CMY(K) models - Gamut

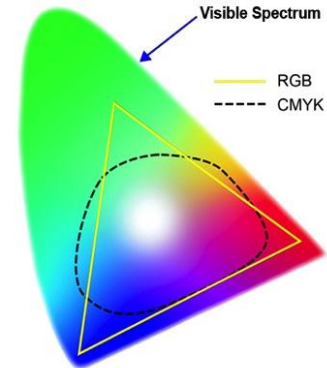
The *gamut* is the range of colors that can be perceived by the human eye

The RGB and CMYK color models can describe only a subset of the whole gamut

Each device has its own gamut: screens, scanners, printers, cameras, ...

Chromatic diagram

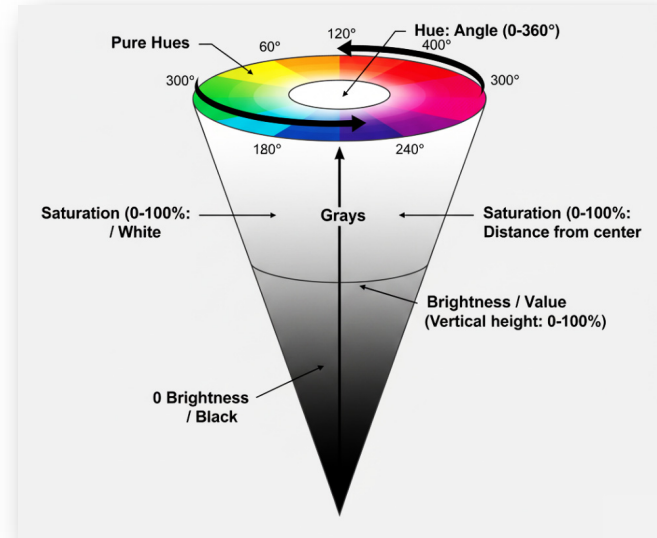
Derives from a “scientific” (mathematical) representation of colors (created by the CIE international authority in 1931 – *Commission Internationale de l'Éclairage*)



HSB/HSV color model...

Color described through three parameters:

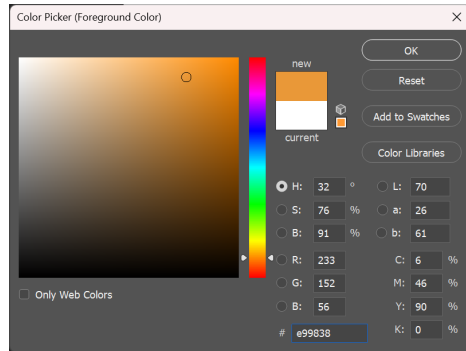
- *Hue*
 - Indicates the tint
 - Values from 0° to 360° on the “wheel of colors”
- *Saturation*
 - Indicates “how much” the color differs from neutral gray
 - Values from 0% (gray) to 100% (full color)
- *Brightness/Value*
 - Indicates how bright the color is
 - Values from 0% to 100%
 - In practice, $B/V = \max(R, G, B)$
(R, G, and B expressed in %)



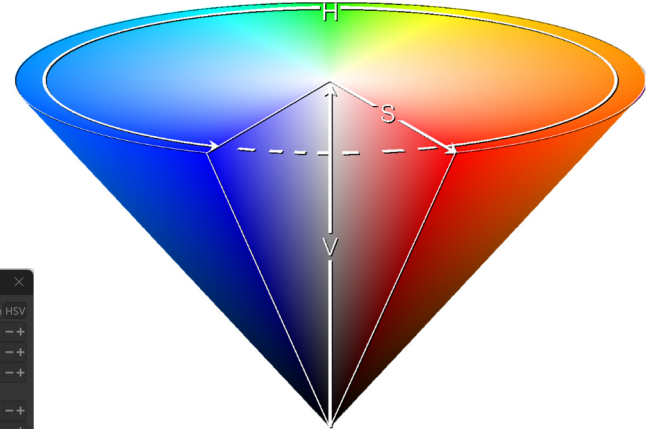
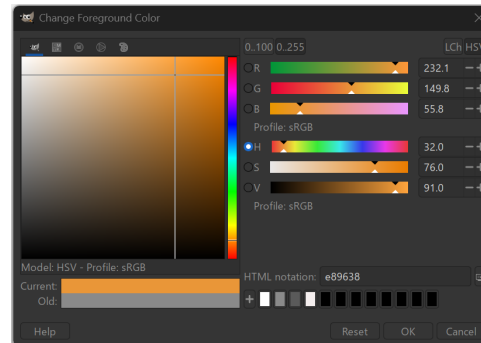
Color coding

... HSB/HSV color model

Photoshop



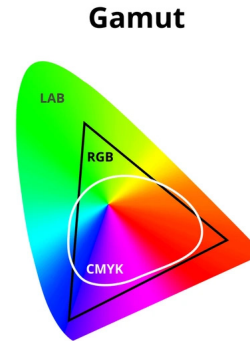
GIMP



CIE models

CIE = Commission Internationale de l'Eclairage

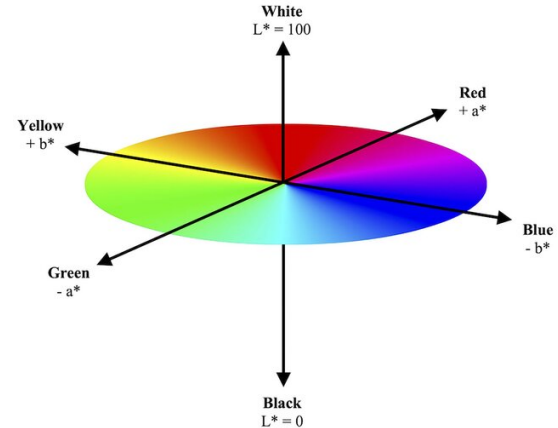
- International Committee created in 1913 to standardize what concerns lighting
- CIE models try to be as independent as possible of specific devices, and are closer to human perception of color
- Three main models:
 - CIE XYZ
 - CIE LUV
 - CIE L^*a^*b



CIE L*a*b model

Three components:

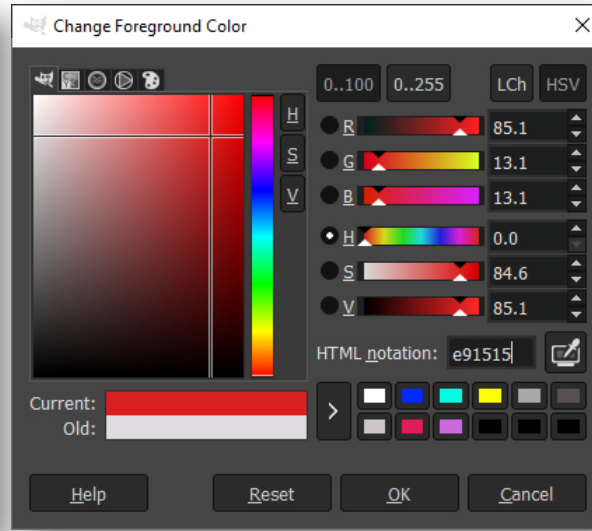
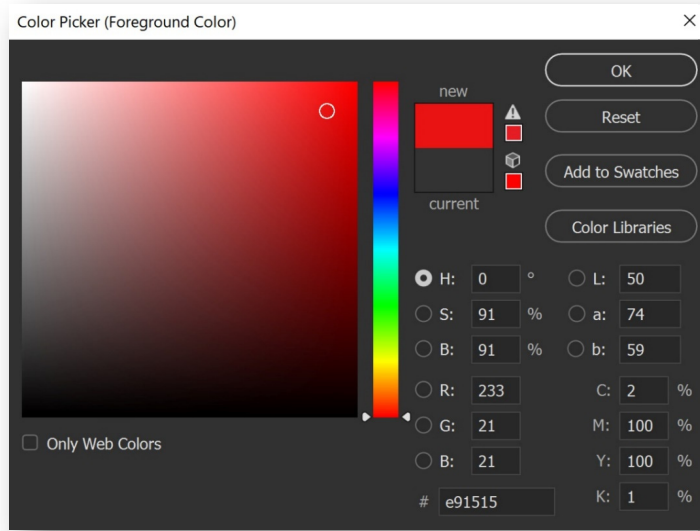
- L (luminance): from 0 (black) to 100 (white)
- a (chromatic component from green to red): from -128 to +127
- b (chromatic component from blue to yellow): from -128 to +127



Color “in practice”

Color selection

E.g., RGB, CMYK, HSB/HSV, and CIE L*a*b models in Adobe Photoshop and GIMP



Color Management System (CMS)...

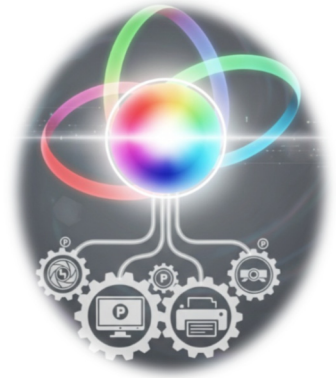
Software that tries to make color representation consistent among different devices

Screens, printers, scanners, cameras, ...

Ideally: screen colors = printed colors

Practically: WYS $\neq$ WYG

Screens use the RGB model, printers use the CMYK model



The CMS exploits a color model that is independent of the device

→ CIE L*a*b model

...Color Management System (CMS)...

Colors are managed within a *virtual color space* (larger than the color spaces of RGB and CMYK)

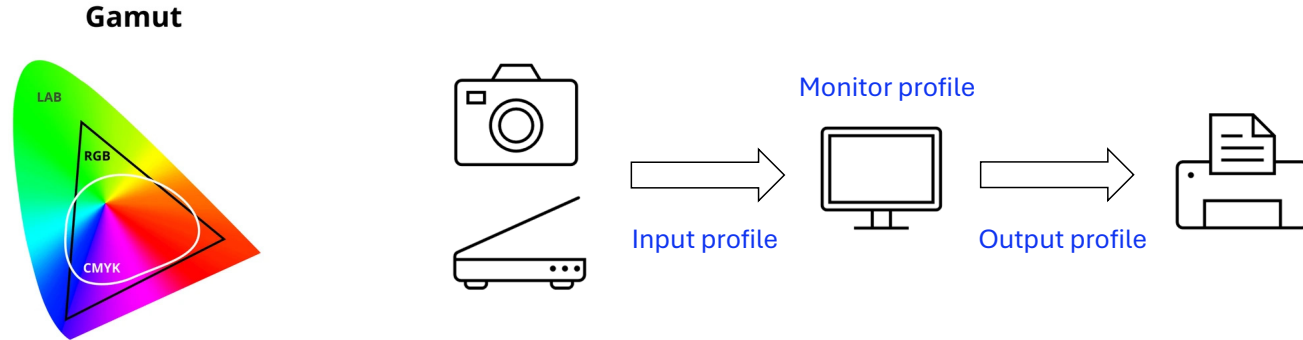
Information about the specific input/output device is stored in its *profile*

ICC CMM

- ICC = International Color Consortium, created in 1993
- CMM = Color Management Model
- The CMS is composed of
 - A color space independent of the device (*Reference Color Space*): usually, CIE L*a*b or CIE XYZ
 - Device profiles
 - A *Color Management Module (CMM)*



...Color Management System (CMS)...



...Color Management System (CMS)...

Device profile

Is a sort of “dictionary”: given specific RGB or CMYK values, it “translates” them into corresponding CIE L^*a^*b values and vice versa

Example 1:

Image acquired with a scanner: a specific color is coded with $R=247$, $G=160$, and $B=91$. The scanner profile specifies that the corresponding L^*a^*b values are $L=76$, $a=19$, and $b=46$. The CMS will convert the L^*a^*b values into the RGB values of the monitor (e.g., $R=250$, $G=175$, $B=100$), using its profile, so that the displayed color will be that of the acquired image

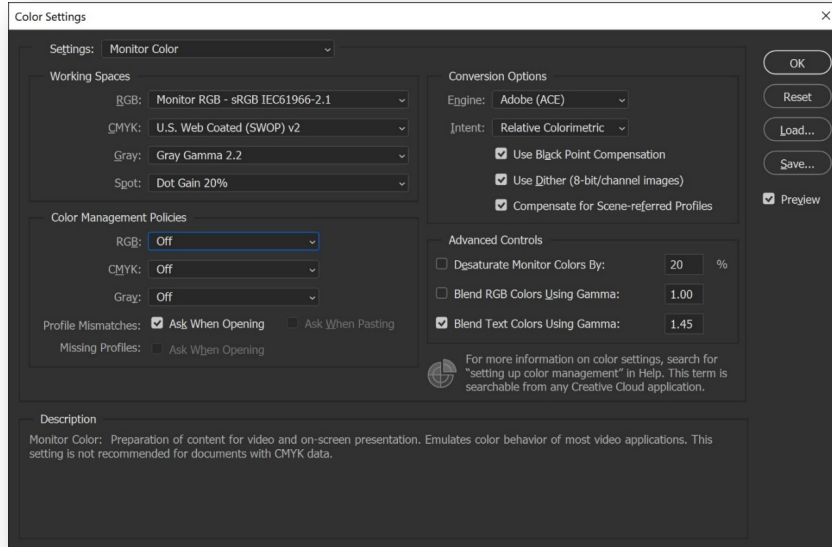
Example 2:

We want to print a picture in which a specific color is coded with the RGB values $R=250$, $G=175$, $B=100$. The CMS will convert them into CIE L^*a^*b values, which, in turn (by means of the printer’s profile) will be converted into the corresponding CMYK values

Color representation

...Color Management System (CMS)

Color profiles in Photoshop and GIMP



Correct screen calibration
is **fundamental**

